

QuickCart Black-Box API Testing Report

Total black-box test cases: 147

Passed: 130

Failed: 17

Skipped: 1

1. Test Design Scope

The black-box suite was designed only from the external QuickCart API documentation. It checks valid requests, invalid inputs, missing fields, wrong data types, and boundary values across the documented endpoints.

- Authentication header validation for X-Roll-Number and X-User-ID.
- Profile retrieval and update validation.
- Address creation, update, delete, and validation rules.
- Product listing, filtering, searching, sorting, and price checks.
- Cart validation, quantity handling, subtotal math, total math, remove, and clear.
- Coupon rules.
- Checkout validation, COD rules, payment statuses, GST, and wallet insufficiency.
- Wallet add-money bounds, pay validation, and exact deduction behavior.
- Loyalty rules.
- Order listing, detail, cancellation, stock restoration, and invoice math.
- Review validation and average-rating behavior.
- Support ticket behavior.
- Admin inspection endpoints used for state verification.

2. Execution Basis

This report is based on the checked-in blackbox/report.html execution artifact, the black-box test suite, and the saved bug notes already present in the repository. A fresh live rerun against <http://localhost:8080> was not possible in this sandbox.

3. Bug Summary

The saved run shows 17 failing test cases, which correspond to 11 distinct logical bugs.

- BUG-01: Cart accepts zero and negative quantities.

- BUG-02: Cart item subtotal calculation is incorrect.
- BUG-03: Cart total calculation is incorrect.
- BUG-04: Checkout allows COD above the documented limit.
- BUG-05: Wallet checkout succeeds with insufficient balance.
- BUG-06: Order cancellation does not restore stock.
- BUG-07: Addresses API accepts invalid letter-based pincodes.
- BUG-08: Profile API accepts invalid phone numbers.
- BUG-09: Reviews API accepts ratings outside 1..5.
- BUG-10: Wallet deduction is not exact.
- BUG-11: Non-existent user IDs return 404 instead of the documented 400.

4. Detailed Bug Notes

- Cart quantity bug: POST /api/v1/cart/add returned 200 OK for invalid quantities 0, -1, and -100.
- Cart subtotal bug: a failing test expected subtotal 360 but observed 104.
- Cart total bug: a failing test expected total 250 but observed 120.
- COD bug: POST /api/v1/checkout allowed COD above the documented limit.
- Wallet checkout bug: checkout succeeded even when the wallet balance was insufficient.
- Stock restoration bug: cancelling an order did not restore product stock correctly.
- Address validation bug: POST /api/v1/addresses accepted pincode abcdef.
- Profile validation bug: PUT /api/v1/profile accepted values such as 12345abcde and abcdefghij.
- Review validation bug: POST /api/v1/products/{product_id}/reviews accepted 0, -1, 6, and 100 as ratings.
- Wallet precision bug: a saved failure log showed expected balance 100800.8 but observed 100800.6.
- Header/status-code bug: non-existent X-User-ID returned 404 instead of 400.

5. Final Blackbox Audit Result

The black-box test suite covers a broad part of the QuickCart API and documents both correct and incorrect behavior. The saved evidence shows many passing validations, but 11 distinct bugs still remain according to the checked-in report artifacts.